



Reinforcement Learning with Verifiable Rewards for Chart Question Answering

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Supervised by

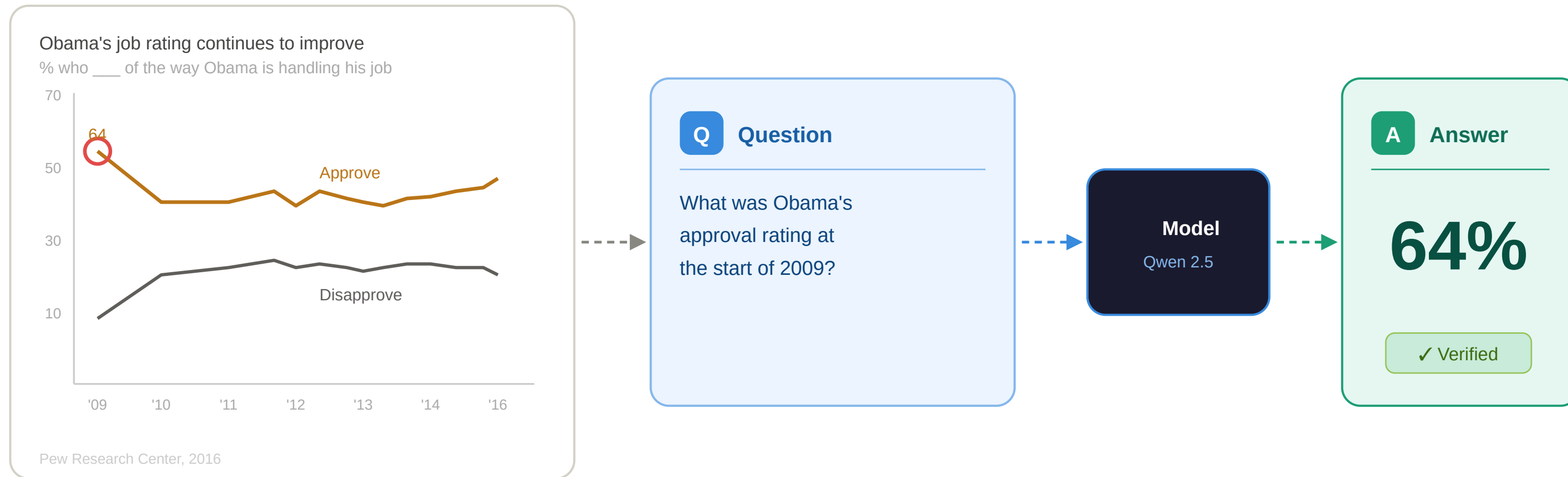
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Chart Question Answering



- ◆ Visual chart understanding task
 - Answer questions directly from chart images
 - Benefits
 1. No manual data annotation needed
 2. Works across chart types & domains

Objectives

- ◆ Fine-tune Qwen 2.5 for chart question answering via RLVR
- ◆ Prevent reasoning collapse on out-of-distribution charts
- ◆ Eliminate hallucinated intermediate reasoning steps
- ◆ Introduce HCPC reward for consistent extraction and diverse reasoning
- ◆ Introduce CLC reward to verify cross-level reasoning coherence
- ◆ Compare GRPO, NSR, and W-REINFORCE as policy update strategies

Problem Statement

Let (I, Q) denote a chart image and a natural language question.

The model learns a mapping:

$$M(I, Q) \rightarrow (R, A)$$

where A is the answer and $R = \{r_1, \dots, r_t\}$ is the supporting reasoning chain.

A response is valid if and only if:

- A is correct with respect to I
- Every step $r_i \in R$ is grounded in values extracted from I

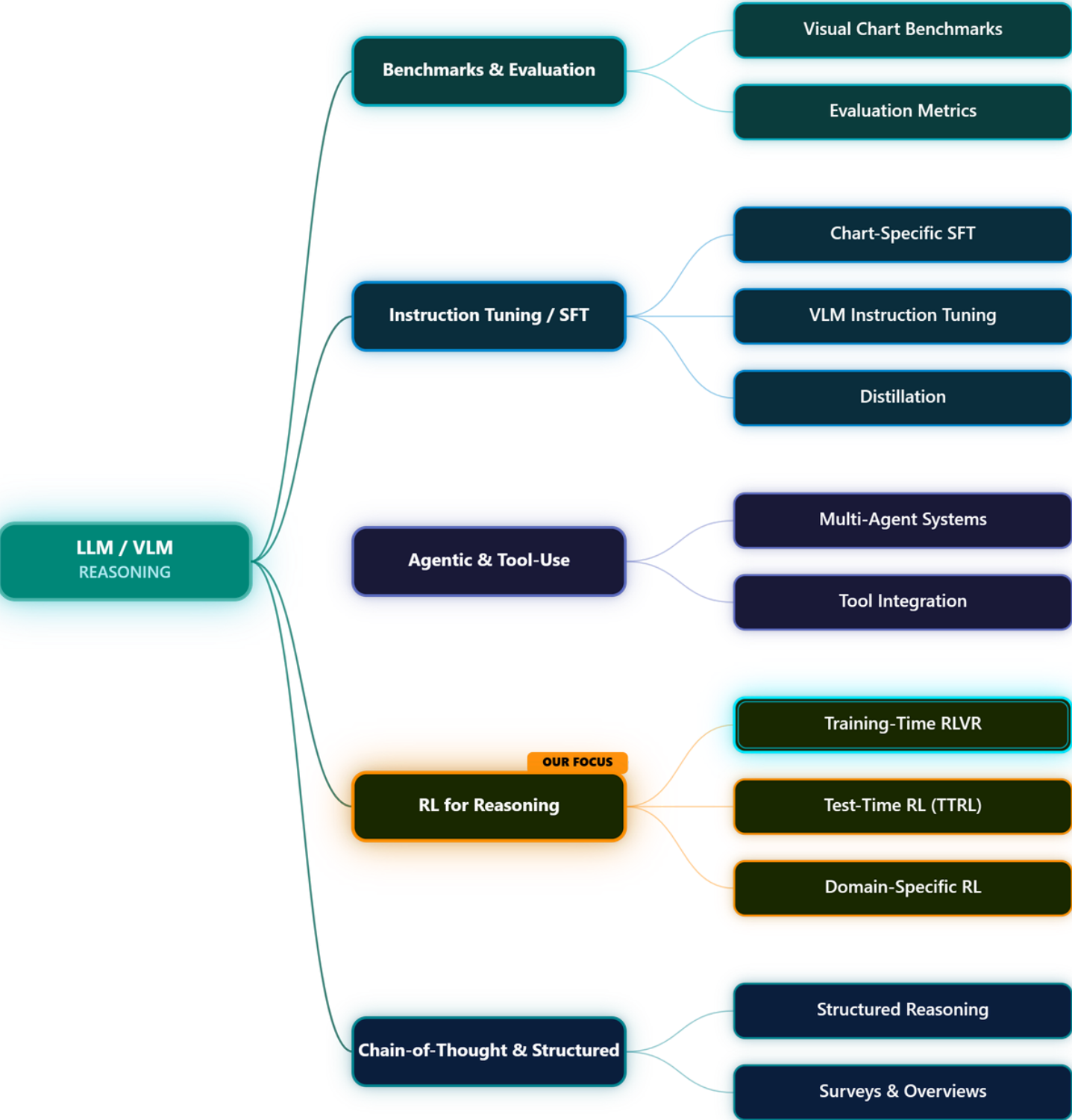
The goal of M is to map a chart image and question to a verifiable answer, supported by a traceable reasoning chain grounded in the chart's extracted values

Background

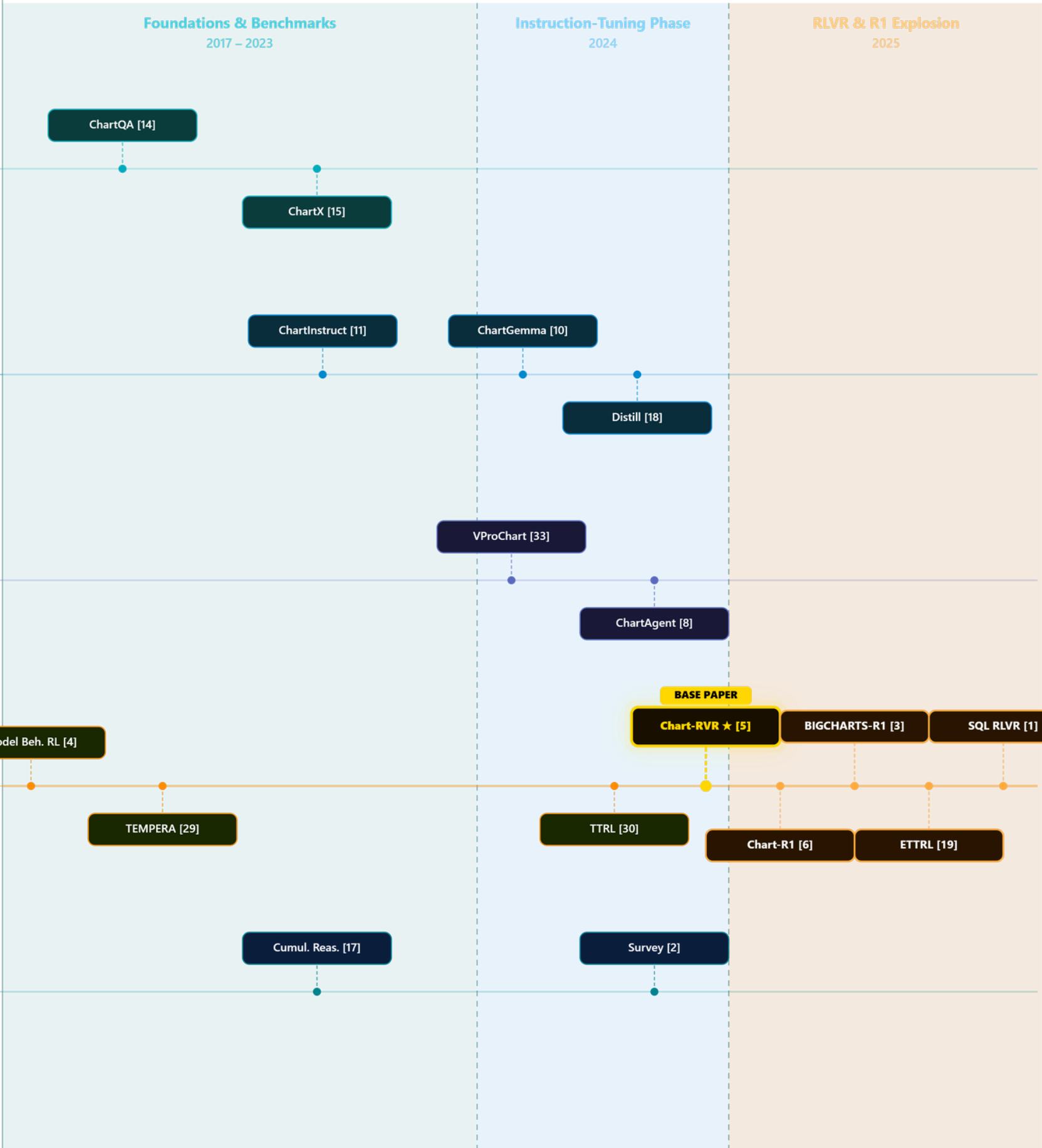
- ◆ Reinforcement learning optimizes a policy by maximizing cumulative rewards through trial and error
- ◆ Rollouts are responses sampled from the model, evaluated and used to compute advantage how much better a response is relative to others
- ◆ RLVR restricts rewards to verifiable, ground-truth correct answers no soft or approximate scoring
- ◆ GRPO updates the policy by strongly reinforcing above-average rollouts and penalizing below-average ones
- ◆ NSR preserves diversity by skipping correct samples entirely and only penalizing wrong ones
- ◆ W-REINFORCE balances accuracy and diversity ,weak positive reinforcement for correct, strong penalty for wrong

Literature Review

APPROACH TAXONOMY



RESEARCH EVOLUTION TIMELINE



Literature Review

Benchmarks, SFT & Agentic Approaches

- **Visual Chart Benchmarks [14] [15] [22] [13]**

- Curated QA pairs over diverse chart types with structured annotations
- Evaluation split into perception (reading) and reasoning (inference) tasks

- **Evaluation Metrics [14] [13]**

1. Relaxed correctness tolerates numeric near-matches
2. Separate scoring for visual grounding vs. logical inference

- **Chart-Specific SFT [11] [7] [12]**

1. chart instruction datasets used to fine-tune VLMs
2. Grounding and attention mechanisms injected

- **Multi-Agent Systems [8] [9] [34] [23]**

- Queries decomposed across specialised planner, extractor, and reasoner agents
- Agent coordination scaled dynamically with test-time compute allocation

- **Tool Integration [33] [28]**

- External solvers and retrievers called to ground numeric and factual answers
- Programmatic reasoning separates chart reading from logical inference

Literature Review

RL for Reasoning & Structured Methods

- **Training-Time RLVR** [5] [6] [3] [26] [20] [16] [1]
 - Rule-based verifiable rewards replace labelled data, enabling reward-driven optimisation
 - GRPO trains chain-of-thought reasoning directly without supervised fine-tuning

- **Evaluation Metrics** [14] [13]
 - Relaxed correctness tolerates numeric near-matches
 - Separate scoring for visual grounding vs. logical inference

- **Domain-Specific RL** [21] [27] [32]

Verifiable reward frameworks transferred to remote sensing, UI navigation, and long-horizon tasks

- **Test-Time RL (TTRL)** [4] [29] [30] [19] [31]
 - Model weights or prompts updated at inference using self-generated reward signals
 - Entropy control balances exploration and exploitation during test-time updates
- **Chain-of-Thought & Structured Reasoning** [17] [25] [24]
 - Sub-step verification and formal symbolic languages enforce interpretable reasoning chains
 - RL-generated traces consistently outperform supervised CoT on complex tasks

Literature Review

Reinforcement Learning with Verifiable Rewards Implicitly Incentivizes Correct Reasoning in Base LLMs

Xumeng Wen ,Zihan Liu ,Shun Zheng, Shengyu Ye, Zhirong Wu, Yang Wang, Zhijian Xu, Xiao Liang, Junjie Li , Ziming Miao, Jiang Bian, Mao Yang *ICLR 2026*

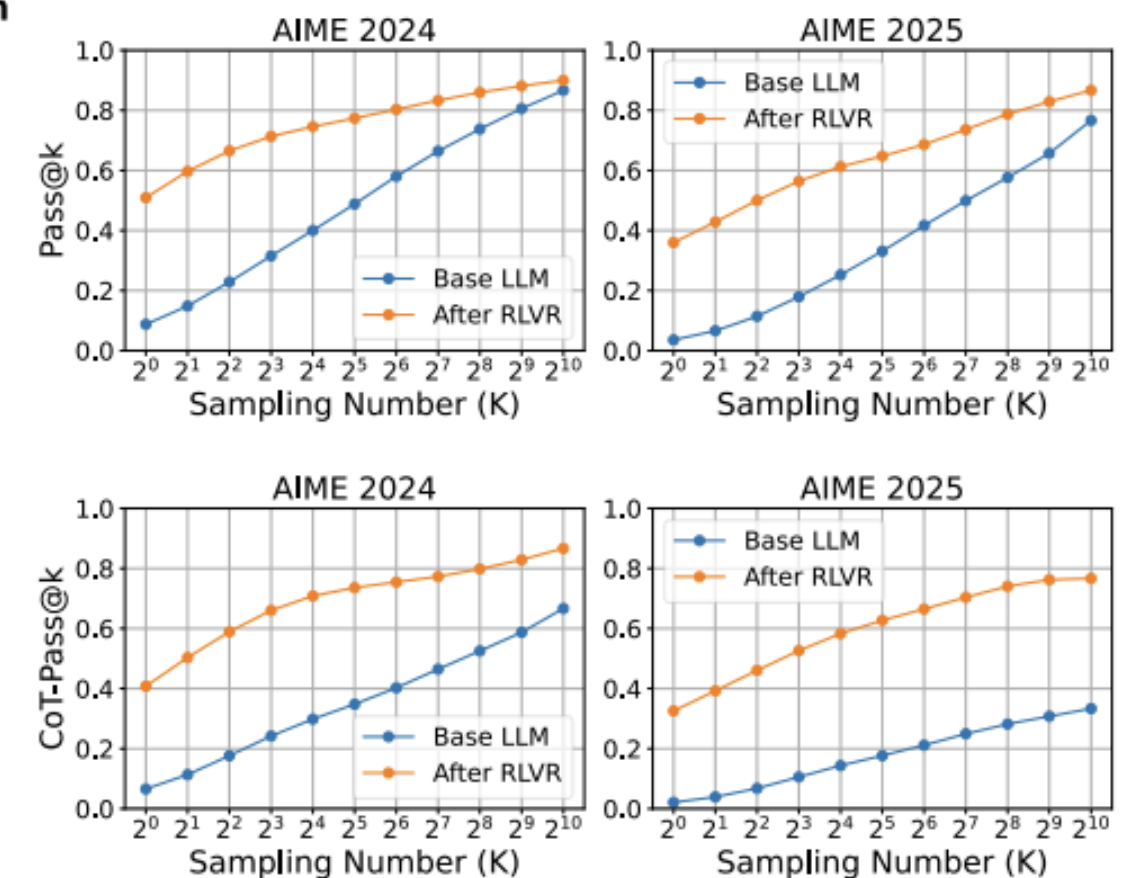
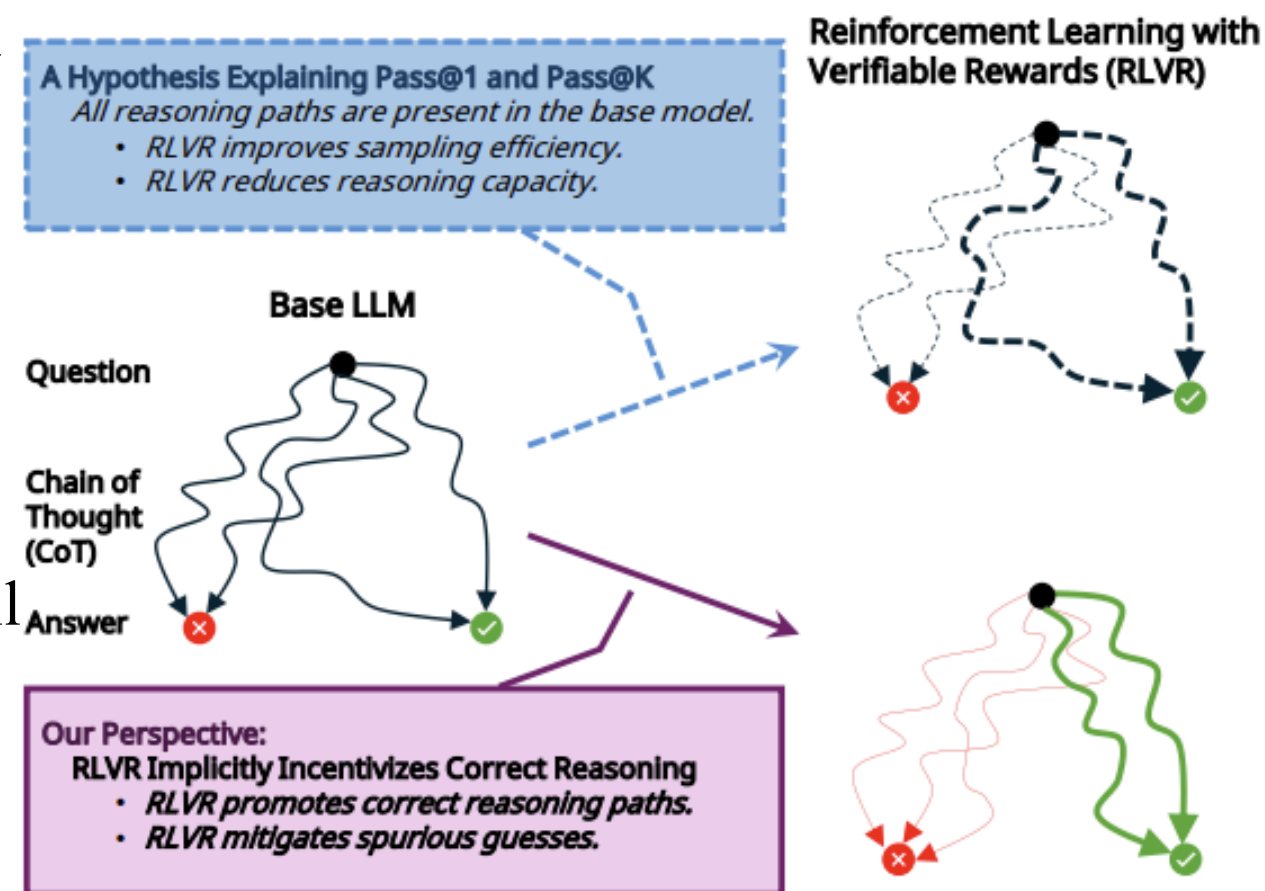
Pass@K fails to capture true reasoning quality, as base models often guess correctly through flawed CoTs. This paper introduces CoT-Pass@K requiring both correct reasoning and answers proving RLVR genuinely improves reasoning, not just sampling efficiency.

Contributions :

- **Introduces CoT-Pass@K** : rewards only correct reasoning + correct answer
- Proves RLVR improves CoT-Pass@K for all K
- Correct reasoning emerges early in training and generalizes to unseen questions
- **LLM-as-a-CoT-Judge** proposed as scalable automated CoT verifier

Drawbacks:

- Doesn't guarantee generalization across all scenarios
- CoT verification relies on an LLM judge.
- Tested mainly on math
- Pass@K degradation at large K reframed, not fixed



Literature Review

Few-Shot Vision-Language Reasoning for Satellite Imagery via Verifiable Rewards

Aybora Koksal, A. Aydin Alatan, ICCV 2025

Introduces the first few-shot RLVR framework for vision-language reasoning on satellite imagery, leveraging rule-based verifiable rewards to achieve competitive performance with minimal supervision eliminating the need for expensive caption annotations.

Contributions :

- First RLVR framework for satellite imagery which does not need caption supervision
- 1-shot training works, a single example yields double-digit gains over the base model
- Generalizes across classification, VQA, and grounding benchmarks

Drawbacks:

- Extreme 1-shot setting causes overfitting
- 8–32 shots needed for stable, broadly transferable gains
- Only tested on Qwen2-VL-2B
- Generalizability beyond satellite imagery unknown

Question: What scene does the image depict? A. Industrial area; B. Farmland; C. Residential area; D. Commercial area. Thinking... The image depicts a large open area with sports fields, which is characteristic of a residential area. The presence of sports fields and greenery suggests a community space for recreation and physical activities. General Knowledge	 Answer: C	Question: Output the bounding box of the following object in the image: a stadium in the middle Thinking: The image shows a stadium with the words \"VISIT FLORIDA\" on its roof, indicating that it is a sports facility in Florida. Visual Grounding	 Answer: [199.0, 155.0, 492.0, 566.0]
Question: Classify the image within one of the given classes: railway station, beach, industrial, center, storage tanks, meadow, mountain, airport, resort, school, port, farmland, medium residential, pond, river, sparse residential, forest, playground, church, parking, viaduct, stadium, square, bridge, desert, dense residential, commercial, park, bare land, baseball field. Thinking... The image shows a large open area with a playground, which is a typical feature of a park. The presence of a playground and the overall layout of the area suggest that this is a park. Scene Classification	 Answer: park	Question: Is there a vineyard? Thinking... There is no indication of a vineyard in the image. The image shows a cityscape with buildings and a river, but there is no visible sign of a vineyard. Visual Question Answering	 Answer: No

Literature Review

Chart-RVR: Reinforcement Learning with Verifiable Rewards for Explainable Chart Reasoning

Sanchit Sinha^{1*}, Oana Frunza², Kashif Rasul², Yuriy Nevmyvaka², Aidong Zhang¹ ¹University of Virginia, ²Morgan Stanley

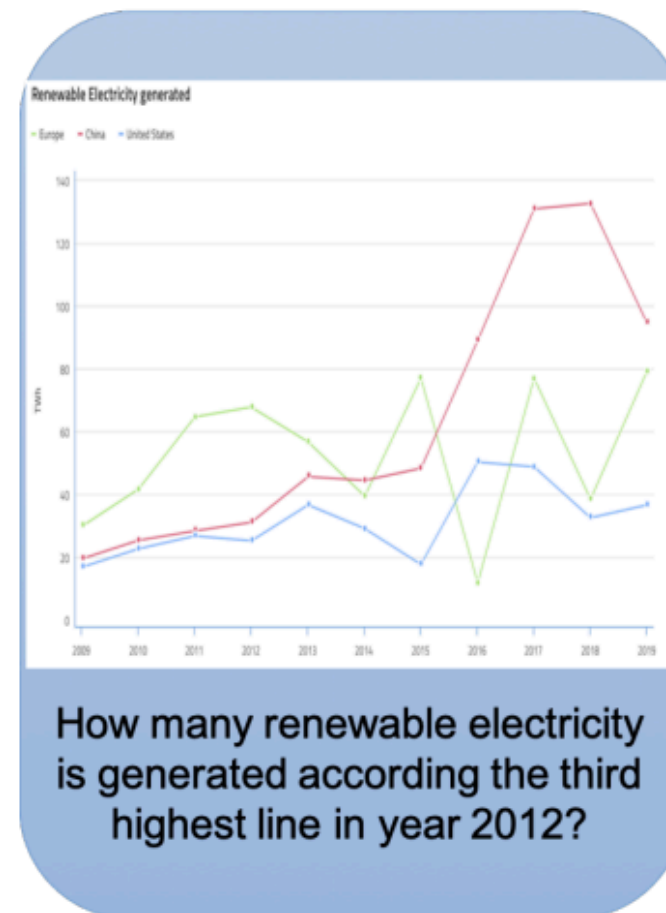
Chart-RVR introduces a GRPO-based reinforcement learning framework with three verifiable rewards : **chart-type classification**, **table reconstruction**, and **process conformity** to improve chart reasoning in vision-language models.

Contribution :

- First general-purpose RL framework with verifiable surrogate-task rewards for chart reasoning
- Outperforms SFT and vanilla GRPO across 3 VLMs (Qwen2.5-VL, Gemma3, InternVL-3.5)
- Closes OOD performance gap, biggest gains on EvoChart, ChartQAPro, ChartBench
- Produces more faithful, interpretable CoT rationales

Drawbacks :

- Limited to 3B-parameter models
- Restricted to chart reasoning
- Requires ground-truth tables and chart types
- Process conformity reward relies on an algorithmic skeleton



CoT

.....

<step-1>First, I identified that the chart is a line graph showing the Renewable Electricity generated over time for three countries..

<step-2>To find the third highest line, I need to compare Europe's value is 68.6, and China value is 32.1

<step-3>Since Europe has the highest value among the three lines in 2012, it cannot be the third highest. Therefore, we need to consider China's as third.

.....

<answer> 32.1 </answer>

SFT

.....

<step-1>: Identify the third highest line in the chart which corresponds to China's data.

<step-2>: Gather the data for China in 2012: 35 TWh

<step-3>: Verify that the value matches the question's requirement.

<step-4>: Confirm there are no discrepancies or errors in the data interpretation.

<answer> 35 </answer>

Chart-RVR

.....

<step-1>: Identify the year 2012, noting the third line, which represents the United States.

.....

<step-3>: Break down the chart, verifying the y-value for 2012, showing 25 TWh for the United States.

<step-4>: Compare the chart, clearly seeing that in 2012, the third line, representing the United States, has a value of 25 TWh.

.....

<answer> 25 </answer>

Literature Review

RLVR-Decomposed: The Surprising Effectiveness of Negative Reinforcement in LLM Reasoning

Xinyu Zhu^{1*}, Mengzhou Xia², Zhepei Wei¹, Wei-Lin Chen², Danqi Chen², Yu Meng¹ ¹University of Virginia, ²Princeton University, *NeurIPS 2025*

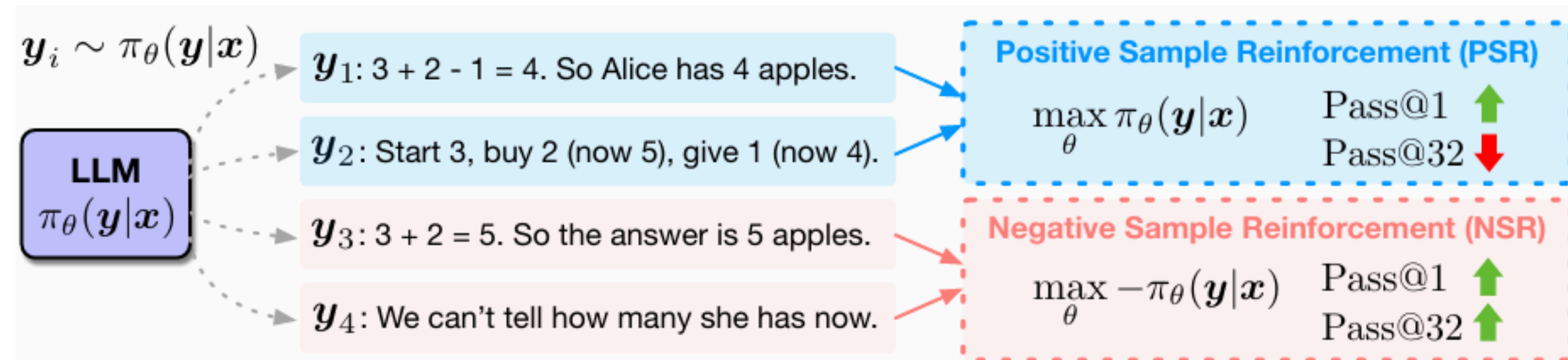
RLVR-Decomposed trains models using reinforcement learning by decomposing the learning signal into positive and negative components, revealing that negative reinforcement alone can be surprisingly effective.

Contributions :

- RLVR decomposed into **two components** PSR and NSR to study their individual effects.
- NSR-only training matches/surpasses PPO & GRPO across full Pass@k spectrum
- NSR preserves diversity by suppressing wrong answers and redistributing probability mass
- Weighted-REINFORCE proposed which upweights NSR, beats PPO & GRPO on MATH, AIME 2025, AMC23

Drawbacks:

- NSR only refines existing knowledge, cannot teach new behaviors
- PSR collapses output diversity, hurts performance at higher k
- Tested only on math



Research Challenges

Collective Disadvantages & Problems Across the discussed Papers

The limitations of the existing models :

- Existing RLVR methods suffer from reasoning collapse and hallucination [**Chart-RVR; CoT-Pass@K**]
- Standard evaluation metrics like Pass@K reward correct final answers regardless of reasoning validity, making RLVR appear ineffective when it is genuinely improving reasoning quality [**CoT-Pass@K; RLVR-Decomposed**]
- Positive reinforcement destroys output diversity [**RLVR-Decomposed; Chart-RVR**]
- Applying RLVR to specialized visual domains demands massive annotated data, making large-scale RLVR training impractical [**Few-Shot RLVR, ICCV 2025; 1-Shot RLVR, NeurIPS 2025**]
- Reward designs treat all reasoning levels uniformly, limiting robustness across visual domains [**Chart-RVR, 2025**]

Research Questions

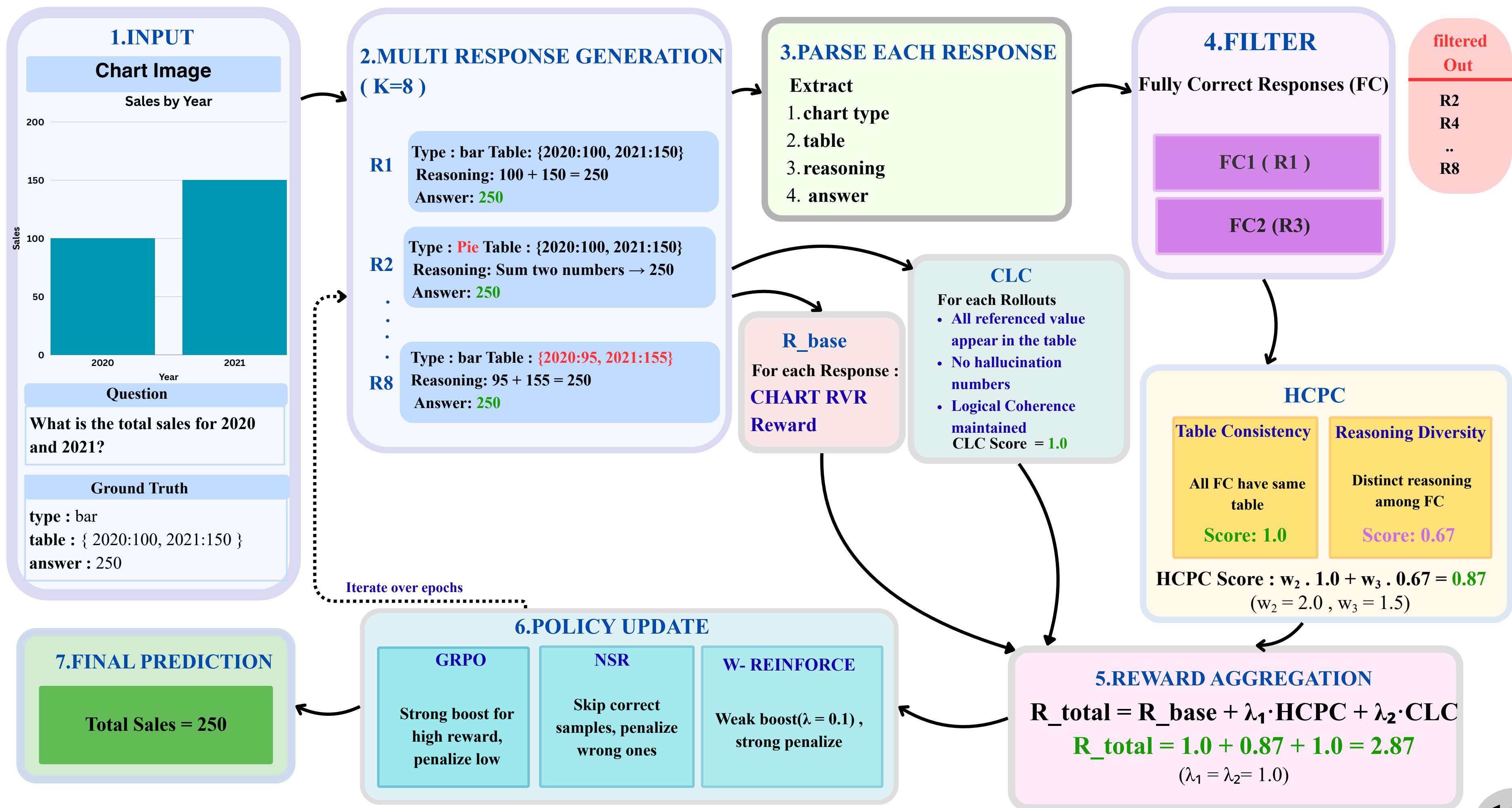
What we are trying to address

- **Can RLVR, with only 1,000 training samples and a 3B-parameter VLM, produce meaningful improvements in chart QA accuracy and output structure?**
 - This would establish whether RLVR is viable for visual reasoning under resource constraints
- **Does a hierarchical reward (HCPC) that conditions bonuses on intermediate reasoning correctness outperform flat GRPO rewards?**
 - Tests whether the path to the answer matters — not just the answer itself
- **Can Negative Sample Reward (NSR), designed for text math, transfer to a vision-language model with multi-component, noisy rewards?**
 - If NSR generalizes to VLMs, it opens a new class of training strategies for visual reasoning beyond our chart task
- **How does explicit diversity promotion (HCPC — directly rewarding diverse reasoning across rollouts) compare to implicit diversity promotion (NSR — preserving diversity by withholding gradients on correct rollouts), in terms of accuracy, rollout reliability, and reasoning quality?**

Proposed Methodology

The components and steps involved in our pipeline :

- **Base Model** : Qwen2.5-VL-3B-Instruct
- **LoRA Fine-tuning** : Parameter efficient method that trains only 4.7M parameters ($r=8$, $\alpha=16$) instead of the full model
- **Rollout Generation** : For each chart-question pair, the model generates 8 candidate responses with moderate randomness
- **Parsing** : Each generated response is decomposed into four parts - predicted chart type, extracted table values, reasoning trace, and final answer.
- **Filtering** : A rollout is kept in the “fully correct” set only if its chart type, extracted table, and final answer all match ground truth, preventing lucky guesses from being rewarded.
- **Rewards** : The total reward consists of three individual rewards
 - **Chart RVR rewards (R_{base})** : Each individual response receives a base reward for correctness and structure, scoring things like format, chart type, table extraction, reasoning quality, final answer, and response length.
 - **Hierarchical Correct-Path Consistency (HCPC)** : A cross-rollout reward that encourages consistent chart-type prediction, consistent table extraction, and diverse reasoning paths among the rollouts that are fully correct.
 - **Cross-Level Coherence (CLC)** : Checks whether the numbers mentioned in the reasoning actually come from the extracted table
- **Policy Update** :
 - **GRPO** : A policy update rule that compares each rollout’s reward to the group average and boosts above-average responses while suppressing below-average ones.
 - **NSR** : A training variant that skips reinforcement on correct samples and only penalizes wrong ones, which helps preserve multiple valid reasoning strategies instead of collapsing to one.
 - **W REINFORCE** : A hybrid update rule that gives weak positive reinforcement to correct responses and strong negative reinforcement to wrong ones, aiming to balance accuracy and diversity.
- **Gradient Update** : Update the model so higher-quality responses become more likely.



Experimental Setup

Coding Environment

Model: Qwen2.5-VL-3B-Instruct

Platforms Used: Kaggle and Local PC

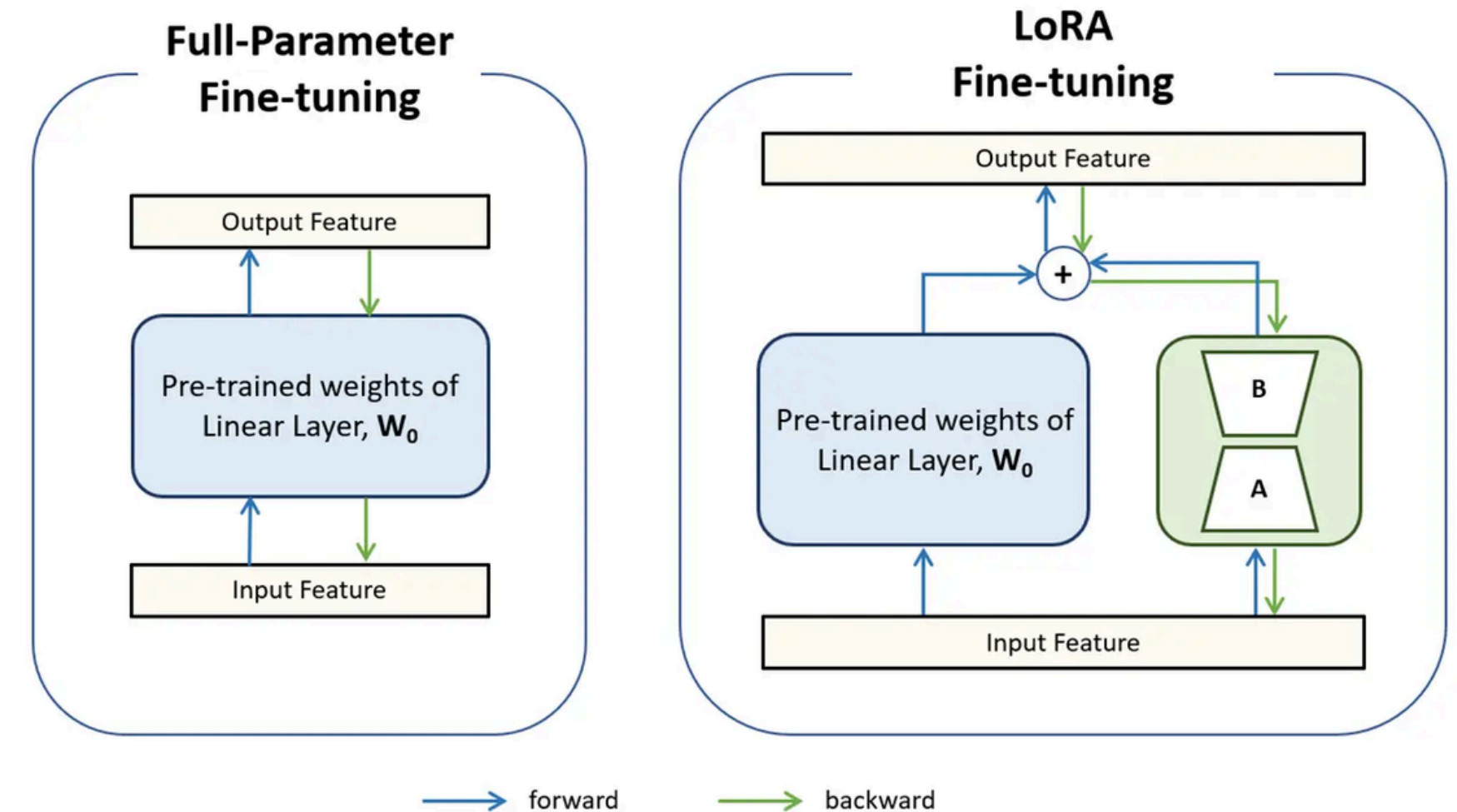
Hardware:

- NVIDIA H100 40GB (Kaggle)
- NVIDIA RTX 3090 24GB VRAM (Local PC)

Frameworks Used:

- transformers + trl — model loading and GRPO training loop
- peft — LoRA adapter integration
- bitsandbytes — bf16 mixed-precision training
- vllm — fast batch inference for evaluation

Training Duration: ~10 hours per method with ~8 hours on evaluation



Experimental Setup

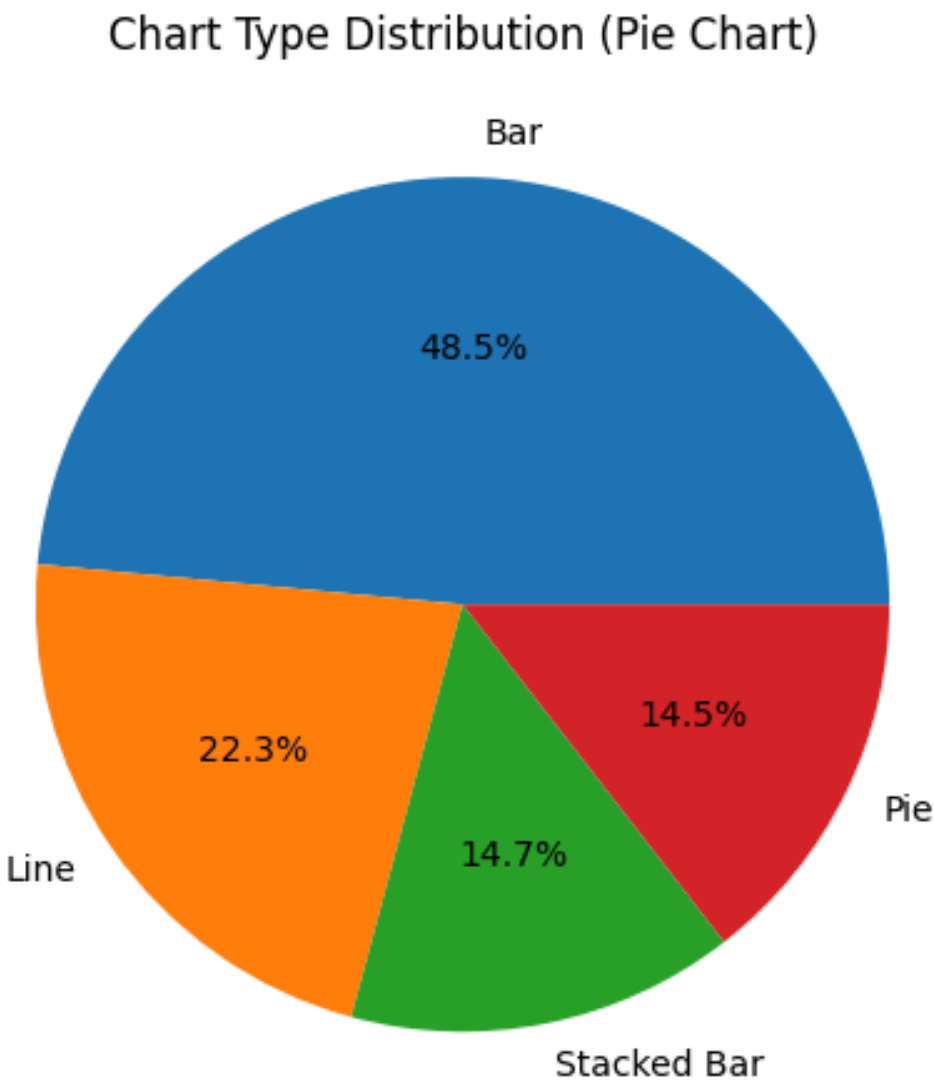
Training Dataset — Chart-RVR GRPO Train [Sinha et al., 2025]

- Full dataset: 34,194 samples (1.78 GB, sourced from ChartQA, PlotQA)
- Subset used: 1,000 samples - random sampling

Test Dataset — ChartQA

- Total Samples Evaluated — 500

Chart Type	Count	Proportion
Bar	485	48.50%
Line	223	22.30%
Stacked Bar	147	14.70%
Pie	145	14.50%
Total	1,000	100%



Experimental Setup

Dataset Samples

System Prompt

You are given a chart image and a query about the chart. Think step-by-step about how to answer the query based on the chart image and then provide the final answer...

Decreasing Japanese satisfaction with functioning of democracy at home

Satisfied with the way democracy is working in Japan



Source: Spring 2018 Global Attitudes Survey. Q4.

PEW RESEARCH CENTER

query

The difference between 2017 and 2018 in the percentage of Japanese satisfaction with democracy is what percentage?

Ground Truth : 10

Output Format

`<type> Bar </type>`

`<table>`

```
{
  "columns": ["Year", "Percentage"],
  "rows": [
    ["2017", "50%"],
    ["2018", "40"] ]
}
```

`</table>`

`<think>`

`<step-1>`: The question asks for

`<step-2>`: From the chart: 2017 = 50%, 2018 = 40%.

`<step-3>`: We need to calculate the difference.

`<step-4>`: The difference is 50% – 40% = 10%.

`<step-5>`: Values taken directly from chart. Verified.

`</think>`

`<answer> 10 </answer>`

Experimental Setup

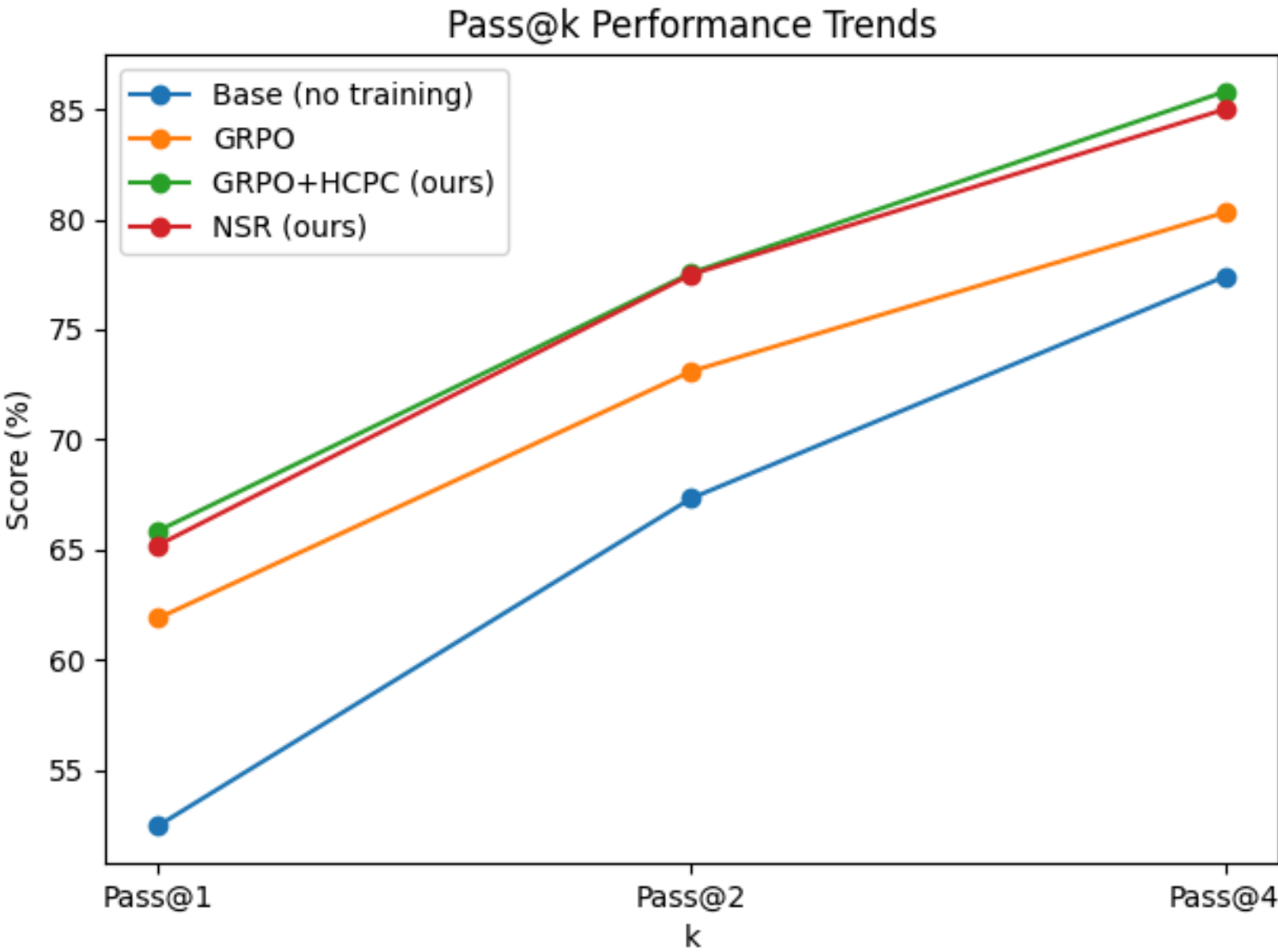
Metrics Evaluated

Metric	Range	What It Measures
Accuracy (Acc%)	0–100%	Fraction of samples answered correctly (relaxed match, at least 1/4 correct)
Pass@K	0–100%	Probability at least 1 of K sampled responses is correct [Chen et al., 2021]
Correct Rate	0.0–1.0	fraction of all individual rollouts correct (total correct / total rollouts across all samples).
D_reason	0.0–1.0	Jaccard distance of reasoning step sets across rollouts — strategy diversity

Results

Performance Evaluation — ChartQA (In-Distribution, 500 samples)

Method	Acc%	Correct Rate	Pass@1	Pass@2	Pass@4	D_reason	C_table
Base (no training)	51.8	52.45	52.45	67.33	77.4	0.04	0.591
GRPO	63	61.7	61.88	73.1	80.31	0.058	0.749
GRPO+HC PC (ours)	62.2	68.4	65.85	77.57	85.8	0.072	0.771
NSR (ours — first VLM)	66.4	67.8	65.2	77.5	85	0.067	0.81



Conclusion and Future Plan

What We Did

- **HCPC (Hierarchical Correct-Path Consistency):** A cross-rollout reward bonus that rewards consistent table extraction and diverse reasoning — conditioned on correctness of intermediate steps, not just final answers
- **NSR applied to VLMs for the first time:** Extends negative-only reinforcement [Zhu et al., 2025] to vision-language models with noisy, multi-component rewards. Used here as a competitive baseline to test whether implicit diversity preservation (withholding gradients on correct rollouts) can match explicit diversity engineering (HCPC).

Future Plans

- **Model scale and recency** — Extending to larger parameter regimes (7B, 72B) and evaluating against more recent vision-language models
- **Full training corpus** — Current results are based on a 1,000-sample subset of the 34,194-sample Chart-RVR dataset.
- **Entropy-based reasoning diversity** — A principled alternative is to directly regularize the entropy of the rollout distribution during training, enabling explicit, continuous control over diversity without dependence on surface-level textual overlap.
- **Out-of-distribution robustness** — Investigate domain-adaptive training strategies and evaluate across a broader range of chart distributions.

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Thank You